

Numerical Analysis - Final Exam

Recalled from memory after the exam · questions only · may contain minor errors

Question 1 (10p)

Find a ring (annulus) on the complex plane that contains all zeros of $p(x) = 2z^5 - 19z^4 + 75z^3 - 156z^2 + 171z - 73$. (Let z_i be a zero of $p(x)$. Find ρ_1 and ρ_2 such that $\rho_1 < |z_i| < \rho_2$ for $1 \leq i \leq 5$.)

Question 2 (20p)

Find the Taylor expansion of the polynomial $p(x) = 2z^5 - 19z^4 + 75z^3 - 156z^2 + 171z - 73$ about the point $z_0 = 2$ using Horner's algorithm.

Question 3 (30p)

Use Gaussian elimination with scaled row pivoting for the matrix A given below to determine a factorization of the form $PA = LU$, where P is the permutation matrix, L is the lower triangular matrix, and U is the upper triangular matrix. (Find L and U .)

$$A = \begin{bmatrix} 12 & -2 & -5 & 28 \\ 6 & 2 & 0 & 92 \\ 6 & -1 & -4 & 13 \\ 3 & -2 & 1 & 4 \end{bmatrix}$$

Question 4

For the function values given in the following table:

x	-2	-1	0	1
$f(x)$	3	5	3	21

- (10p) Find the Lagrange form of the interpolating polynomial.
- (10p) Find the Newton form of the interpolating polynomial.

Question 5

Assume that we interpolate the function $f(x) = e^{6+2x}$ with a polynomial $p(x)$ of degree 5 using 6 nodes in $[-1, 1]$.

- (10p) If we choose the nodes arbitrarily, what will be the upper bound for $|f(x) - p(x)|$ in $[-1, 1]$?
- (10p) What will be the node values to make the upper bound minimum for $|f(x) - p(x)|$ in $[-1, 1]$?
- (10p) What will be the minimum upper bound for $|f(x) - p(x)|$ in $[-1, 1]$?